

Technical Challenges, Public Policy and Use-cases (Smart Contracts/ DeFi)

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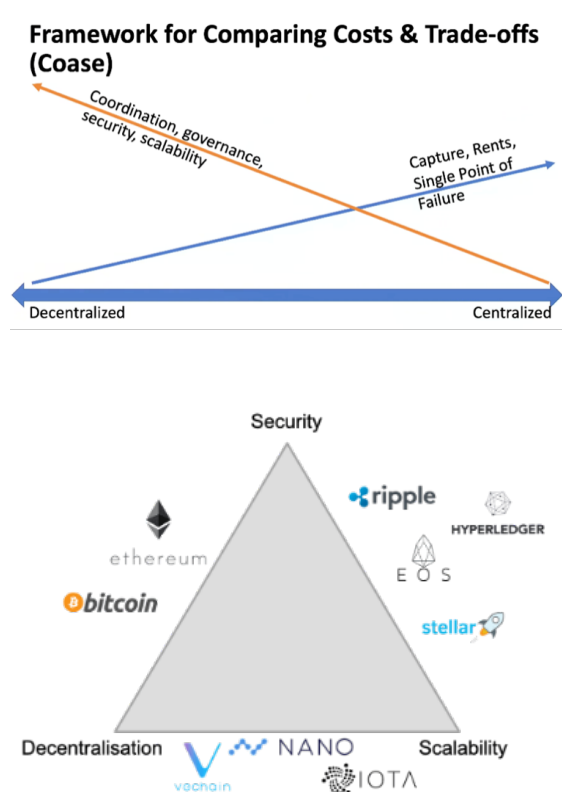
Technical Challenges

Many cryptocurrencies have the same fundamental building blocks:

- Cryptography & Timestamped Logs
- Decentralized Network Consensus
- Transaction Code & Ledgers

However, they all approach different solutions to solve the Trilemma issue, coined by Vitalik Butlerin (Founder of Ethereum).

The trilemma issue highlights the difficulty in incorporating Scalability, Security and Decentralisation, where the theory/issue states you can only have 2 of the 3.



The likelihood of us cracking the Trilemma will only grow as more smart minds come together to look for solutions.

Scalability

As seen from the triangle above, BTC has greater focus on security and decentralisation and therefore focuses on scalability. These issues are highlighted by the figures below:

BTC's Layer 1 Throughput: 7-10 tx/s

ETH's Throughput: 16- 20tx/s

Visa's Throughput: 24000tx/s

Layer 2's can increase the transactions significantly

- BTC uses the Lightning Network
- ETH focuses on the like of Zero Knowledge (ZK)-roll ups
 - Essentially users migrate to Layer 2's to perform transactions, yet the security and decentralisation of the main-net comes from the original layers

Layer 2 / Side chains draw parallels to current financial system: Central banks manage the overarching tx's (like BTC layer 1), then middle banks roll up transactions and send to layer one (like side chains): Lots of activity → Rolled into side-chains → Broadcast to main chain

Another contributor to slow networks is: Bigger blocks → heavier blockchain (more sluggish) → and more processing power needed to support → less nodes → less decentralised.

ETH's Change to Proof of Stake

In September 2022, ETH proceeded with a fork, called the merge, where they shifted ETH's consensus mechanism to POS, allowing for nodes, given they stake a certain value of currency. This did not necessarily change the transactions/second (tps) but had other benefits which help the long-term sustainability of ETH.

- Cut energy use by 99.9%.
- Improve block finality (12 seconds per slot, 2 epochs (12 min) for finality).
- Lay groundwork for scaling upgrades like Danksharding / Proto-Danksharding (EIP-4844), which massively reduces L2 fees.

Other currencies, like Solana for example use different consensus mechanisms to navigate the trilemma issue but forgo decentralization. For many Crypto Punks, decentralization is key.

Also, XMR (a PoW CC) solves the trilemma issue, operating as well functioning cash. However, its extreme focus on privacy has cast doubts on it from a public policy perspective.

Alternative Consensus Protocols

Generally Randomized or Delegated Selection of Nodes to Validate next Block

- May have added mechanism to confirm Block Validators' Work

Randomized Selection May be Based upon:

- Proof of Stake – Stake in Native Currency
- Proof of Activity - Hybrid of POW and POS
- Proof of Burn – Validation comes with Burning of Coins
- Proof of Capacity (Storage or Space) – Based upon Hardware Space

Delegated Selection May be Based upon Tiered System of Nodes

Major Permissionless Blockchain Applications still use Proof of Work – though:

- DASH is a hybrid of POW with a tiered system of 'Masternodes'
- NEO uses a Delegated protocol of 'Professional Nodes'



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Public Policy

Governments have a vested interest in the safety of its citizens, guarding against illicit activities and financial stability. The newcomer, of blockchain and cryptocurrencies, provides an avenue for these issues to flourish, and hence present the need for public policy controls.

- **Guarding Against Illicit Activity**

- Risks: tax evasion, money laundering, dark markets, pseudonymous addresses, privacy coins, mixers, decentralized exchanges.
- Consensus: broad agreement that Anti-Money Laundering (AML)/Know Your Customer (KYC) and compliance frameworks are needed, but global implementation remains uneven.
- Difficulty of Taxation: ps. most treat crypto as property; mining taxed as income.

- **Protecting the Investing Public**

- Goals: prevent fraud/manipulation, ensure fair disclosure, safeguard custodial arrangements, promote transparent secondary markets, and manage conflicts of interest.
- Initial Coin Offerings (ICOs, the CC version of IPOs) often meet securities test criteria (Howey Test) → many regulators (e.g., U.S., Canada) push to treat ICOs as securities.
- Investor protection underpins trust, adoption, and long-term economic benefits.

- **Financial Stability**

- Current risk small relative to global markets, but systemic risk could grow with leverage and wider adoption.
- Levels of concern vary: capital-control economies (e.g., China) fear capital flight, while others see innovation potential.
- Blockchain's decentralization may enhance resilience compared to traditional centralized financial systems (No one messes up monetary policy)
 - * Legacy systems → Lots of data → opportunity for blockchain tech
 - * Economic rents (2.5%) → Can reduce this through centralisation (2.7% on payments)
 - * Concentrated risks → Banks are isolated. . .
 - * Infrastructure Systems' costs & counterparty risks
 - * Periodic crisis and instability
 - * Financial inclusion

- **Global Policy Divergence**

- Some countries encourage innovation; others move to integrate crypto within securities laws.
- Consensus unlikely due to differing economic histories, inflation experiences, and regulatory philosophies.
- Mark Carney: crypto assets are “not currencies” but volatile assets; authorities must choose to isolate, integrate, or regulate.

Privacy

Cryptocurrencies raise fundamental questions about privacy:

- **Too much privacy (government concern):**

- Governments and regulators worry that strong anonymity can **reduce the tax pool, undermine financial transparency, and facilitate illicit activity** (e.g., money laundering, terrorist financing, ransomware payments).
- Privacy coins such as Monero and Zcash, which use advanced cryptographic techniques like mixers, tumblers, or zero-knowledge proofs (ZKPs), are often highlighted as potential enablers of crime.
 - * However, it is also a fundamental right, and the backbone of BTC/Crypto Punk movement
 - * Governments like the idea of CBDC’s (Big brother society)

- **Too little privacy (corporate/individual concern):**

- On the other hand, blockchains like Bitcoin and Ethereum are pseudonymous but transparent: all transactions are visible to anyone.
- This creates risks of financial surveillance by corporations, data miners, or even authoritarian states.
- For ordinary citizens, this can mean the loss of financial privacy, which is increasingly recognized as part of the broader framework of human rights (similar to data privacy).

Thus, public policy must navigate between preventing abuse and protecting individual freedom. The technology within privacy coins can however also be the source of the issue as well as the solution:

Zero-Knowledge Proofs (ZKPs):

- ZKPs allow a user to prove a statement is true without revealing underlying details.
- In crypto, this could mean proving you have sufficient funds to spend without disclosing your balance or transaction history.

- This technology creates the possibility of **privacy-preserving compliance**—allowing oversight without mass exposure of personal data.

Pseudonymity vs. Anonymity:

- Public blockchains (e.g., Bitcoin) are pseudonymous, meaning addresses are not linked to identities by default but can often be traced through blockchain analytics.
- Privacy coins such as Monero aim for full anonymity, complicating law enforcement.

In essence, cryptocurrencies challenge traditional notions of privacy, security, and regulation.

- Too much privacy risks undermining law and order; too little privacy risks eroding individual freedoms.
- Public policy will need to evolve beyond simple prohibition or blanket surveillance.
- The most promising path lies in privacy-preserving compliance frameworks enabled by cryptography (e.g., zero-knowledge proofs), where both regulators and users can achieve their goals.

Interoperability of Blockchains

One of the central barriers to mainstream blockchain adoption is interoperability—the ability for different systems to interact seamlessly:

- **Fiat–Crypto Linkages:** Bridging blockchains with legacy financial networks introduces frictions, higher costs of trust, and complex compliance burdens. Without reliable interoperability, value transfer between fiat and crypto remains slow, fragmented, and prone to risk.
- **Cross-Chain Communication:** Different blockchains often operate as closed ecosystems. Without decentralized mechanisms (e.g. cross-chain protocols, oracles - such as Chainlink), assets and data cannot flow securely between chains, limiting scalability.
- **Multiple Cryptocurrencies:** The lack of standardized frameworks means exchanges, wallets, and payment processors must act as intermediaries, raising costs and creating regulatory blind spots.

Policy Angle: Governments should incentivize the development of secure, standardized interoperability frameworks—balancing innovation with oversight. Clear standards could support financial integration while reducing systemic risk.

Governance

Blockchain governance determines how networks evolve, who makes decisions, and whether assets are treated as commodities (like gold) or securities (like stocks).

- **Consensus and Upgrades:** Because software updates require broad consensus, disagreements often result in hard forks (splitting into new chains), undermining stability. This complicates regulatory classification: assets behaving like securities require investor protections, while “store of value” tokens may be treated as commodities.
- **Permissionless vs. Permissioned Systems:** Public blockchains allow open participation but struggle with coordination, whereas financial institutions prefer permissioned systems with clear governance—closer to existing regulatory norms.
- **Policy Trade-Off:** Overly rigid classification risks stifling innovation, but too much flexibility creates legal uncertainty for investors and developers.

Policy Angle: Regulators should define governance standards that clarify when a token is a commodity vs. a security, while encouraging transparency in upgrade processes. A balanced approach would reduce legal risk, attract institutional adoption, and maintain blockchain’s open innovation potential.

Classification as Commodity or Security

- How cryptocurrencies are classified has profound regulatory consequences. If defined as a commodity, they fall under market integrity and trading rules, emphasizing fraud prevention and transparency
- If defined as a security they must comply with stricter securities laws, including disclosure obligations, investor protections, and restrictions on issuance and trading
 - This distinction shapes taxation, exchange regulation, and the legal framework for Initial Coin Offerings (ICOs).
 - Corporate Structure Security (XRP)
 - Decentralized Commodity (ETH/BTC)

Introduction to use-cases

Blockchain Based Use-Cases / Benefits

Blockchain Economics affects certain key points:

1. Verification – Lowers cost of verification
 - (a) Direct Costs: No back-ends/back offices to ensure verification.

- (b) Privacy and Data leakage costs: Don't exist due to pseudonymity.
 - (c) Censorship Risks:
 - i. Distributed decentralised blockchains are more censorship resistant. No central authority privy to information/decisions.
 - (d) Settlement – Timeliness and Certainty of Finality:
 - i. All chargers pay 2.5%, many merchants don't like this. BTC is a solution.
 - (e) Costs of Trust:
 - i. Trust in code vs. central intermediary.
 - (f) Economic rents due to market power.
2. Networks – The value of moving property rights:
- (a) Blockchain can reduce developer costs: Provides source to create a network
 - (b) Tokens incentivise and help fund networks: New form of crowd funding
 - (c) Incentive mechanism during an operating phase

Financial Sector Potential Use Cases

- **Venture Capital – ICOs**

- **What:** Raise funds via Initial Coin Offerings.
- **How:** Token sales replace traditional equity/VC routes.
- **Why:** Lower entry barriers, global investor access, fast liquidity.

- **Payment Systems**

- **What:** Use blockchain rails for payments (cross-border, interbank, retail).
- **How:** Near-instant settlement without intermediaries.
- **Why:** Reduce fees, speed up remittances, increase transparency.

- **Loan Issuance & Trade Finance**

- **What:** Smart contracts replace paper-based loan/trade documents.
- **How:** Automate verification, collateral management, repayment.
- **Why:** Cut paperwork, fraud, and delays in financing global trade.

- **Clearance, Settlement & Processing**

- **What:** Apply Distributed Ledger Technology (DLT) to securities/derivatives markets.
- **How:** Real-time clearing and atomic settlement of trades.

- **Why:** Reduce counterparty risk, improve efficiency, lower back-office costs.
- **Digital IDs & Data Reporting**
 - **What:** Blockchain-based self-sovereign IDs.
 - **How:** Encrypted identity stored on chain, controlled by users.
 - **Why:** Enhance privacy, prevent fraud, streamline compliance (KYC/AML).
- **Central Bank Digital Currencies (CBDCs)**
 - **What:** State-backed digital currency on a blockchain-like system.
 - **How:** Issued by central banks for wholesale or retail use.
 - **Why:** Modernize monetary systems, improve policy control, rival private stablecoins.

Smart Contract / DeFi / NFT Use Cases

Smart contracts are the building blocks that make decentralised systems like DeFi and NFTs possible. In simple terms, a smart contract is a **self-executing piece of code** that lives on a blockchain. It runs automatically when certain conditions are met, and no one can change or interfere with it once it is deployed.

Think of a smart contract as an **automated agreement**. For example, you might create a contract that says: “If someone sends 1 ETH to this address, automatically send them back 100 tokens.” Once this contract is on the blockchain, it will execute that rule every time, without needing a person to approve the transaction.

Smart contracts eliminate the need for a **trusted third party**. In traditional finance, if you wanted to send money to someone in exchange for a service, you’d likely go through a bank, a payment processor, or a legal contract. With a smart contract, the terms are enforced by code. The participants don’t need to trust the each other— you just need to trust that the code does what it says.

These contracts are **immutable** — once deployed, they can’t be altered unless they are specifically designed to be upgradeable. This gives users confidence that the rules won’t change later.

Why smart contracts matter:

- They automate complex processes without human involvement.
- They allow people who don’t know each other to interact and transact securely, without needing to trust one another.
- They power nearly every major Web3 use case including:
 - NFT marketplaces

- DeFi protocols like Uniswap and AAVE
- DAOs for decentralised governance
- Gaming mechanics.

Analogy:

A smart contract is like a **vending machine**. You insert the correct amount of money and press a button. If the amount is correct and the product is in stock, the machine automatically gives you what you asked for and no cashier needed. If the amount is wrong, nothing happens. It simply enforces the rules without emotion or judgment.

This is how smart contracts work. They hold funds, check for conditions, then fulfil the purpose of the contract; all without needing a bank, lawyer, or manager to be involved.

Turing-Complete Smart Contracts: Why BTC doesn't have smart contracts

- Turing-complete scripting means a programming language can perform any computation given enough resources (loops, conditionals, storage).
- Ethereum and many Layer 2s support Turing-complete languages like Solidity or Vyper, enabling fully programmable contracts that can encode complex logic for payments, lending, trading, or conditional interactions.
- Bitcoin scripting is not Turing-complete; it's intentionally limited for security, supporting only basic operations like signature verification and simple conditional transfers—hence, complex smart contracts aren't natively possible.
- Layer 2 & DeFi Execution:
 - **Layer 2 solutions** (e.g., Optimism, Arbitrum) run on top of Ethereum or other chains, bundling transactions in rollups or sidechains. This reduces costs and increases speed, making high-volume smart contract interactions feasible.

DeFi Coins and How DeFi Challenges Traditional Finance

Decentralised Finance, or DeFi, refers to financial services that are built and run on public blockchains like Ethereum. These services are designed to function without banks, brokers, or traditional financial institutions. Instead of using human-run systems or centralised platforms, DeFi platforms rely on code (smart contracts) to carry out lending, borrowing, trading, saving, transferring money, calculating interest, and other financial functions.

The goal of DeFi is to replace or reimagine traditional financial infrastructure in a way that is open, transparent and accessible to anyone regardless of their country, income level or banking status.

Since everything runs on-chain:

- There's no need for user applications to be reviewed
- No banks to approve a loan
- No delay in processing transactions
- If you meet the conditions (e.g. have the collateral), you can use the platform

In DeFi, all you need is a crypto wallet like OpenWallet and an internet connection. No one can stop you from accessing these services. This is what we mean when we say DeFi is permissionless there's no sign-up process or human gatekeeper deciding whether you're allowed to participate.

Key benefits of DeFi compared to traditional finance:

- There are no middlemen: You don't need a bank, a broker, or a loan officer. Instead, you interact directly with smart contracts. This reduces overhead and often results in lower fees.
- Global access: Anyone in the world with a crypto wallet can access the same services. You don't need to live in a rich country or have a bank account to borrow or lend.
- Always open: DeFi never closes. Platforms are live 24/7, with no holidays, weekends, or opening hours.
- Full transparency: All transactions are recorded publicly on the blockchain. Anyone can verify how a platform works, how funds are handled, and where money moves. Smart contract code is often open source.

Some well-known DeFi platforms and tokens:

- AAVE – A decentralised lending protocol. Users can deposit crypto to earn interest or borrow against their assets. Borrowing is over collateralised, meaning you must deposit more than you borrow to protect the system from risk.
- Uniswap – A decentralised exchange (DEX) that allows users to swap 'ERC-20' (Ethereum Request for Comment) tokens directly from their wallets. It uses automated liquidity pools instead of order books.
- Compound – Another lending protocol that automatically adjusts interest rates based on supply and demand. Users can earn yield by supplying assets or borrow against their holdings – CC equivalent of bonds (Provide liquidity to lending pools)

- SushiSwap - (OpenWallet uses this) Originally a community fork of Uniswap, it added extra incentives for liquidity providers and integrated more DeFi tools under one interface.
- Synthetix – A protocol that lets users create and trade synthetic assets i.e tokens that track the price of real-world assets like USD, gold, or stocks.

Each of these platforms is governed by a token (like AAVE, COMP, or SNX), which is often used for voting on decisions or collecting a share of protocol fees.

DeFi Services:

- **Decentralized Exchanges (DEXs):** Peer-to-peer token swaps via liquidity pools instead of centralized order books.
- **Staking & Yield Farming:** Users lock tokens into contracts to earn rewards or interest, incentivizing liquidity provision.
- **Other Services:** Lending, borrowing, synthetic assets, insurance, and automated yield optimization—all executed automatically by smart contracts.
- **Current & Future Potential Use Cases:** Cross-border payments, tokenized real-world assets, decentralized insurance, NFT marketplaces, supply chain verification, decentralized identity management, and programmable financial instruments.

NFTs (Non-Fungible Tokens) – Quick Introduction

- NFTs are unique digital assets recorded on a blockchain, representing ownership of art, collectibles, music, or other digital content.
- Unlike cryptocurrencies (BTC, ETH), NFTs are non-fungible, meaning each token is distinct and cannot be exchanged 1:1 with another.
- They rely on smart contracts for provenance, ownership verification, and transferability.
- Use cases: digital art, gaming items, virtual real estate, collectibles, and tokenized real-world assets.

They essentially provide a way to digitally prove ownership and authenticity of unique items.